YEAR 7 MATHS · CHAPTER 10 · MIND MAP

Measurement

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1 Metric units of length

1 km = 1000 m, 1 m = 100 cm, 1 cm = 10 mm.

2.5 m = **250 cm**

3400 m = **3.4 km**

45 mm = **4.5 cm**

Big unit to small: multiply. Small unit to big: divide.

2 Perimeter

Add all the sides.

Rectangle 7 by 4: $2(7+4)=22$

Square side 5: $4\times5=20$

Regular hexagon, side 3: $6\times3=18$

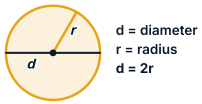
3 Circles and circumference

$C=\pi d=2\pi r$. The radius is half the diameter.

$d=6$: $C=6\pi\approx18.8$

$r=4$: $C=8\pi\approx25.1$

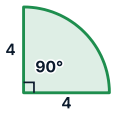
$\pi\approx3.14$



4 Arcs and sectors

Arc length $=\frac{\theta}{360}\times2\pi r$. Add the straight edges for the perimeter.

Quarter circle $r=4$: arc $2\pi\approx6.3$, perimeter $8+2\pi\approx14.3$



5 Area of rectangles

$A=lw$, in square units.

$8\times3=24$ cm²

1 cm² = 100 mm², 1 m² = 10 000 cm²

Square, side 5: 25 cm²

2 m by 3 m: 6 m²

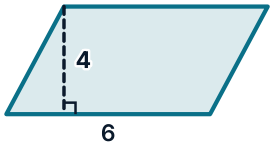
Watch out: to convert area units, square the length conversion.

6 Area of parallelograms

$A=bh$, with the height at right angles to the base.

Base 6, height 4: $A=24$

Base 8 cm, height 5 cm: $A=40$ cm²

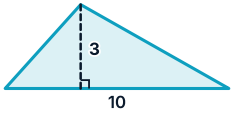


Watch out: use the perpendicular height, not the slanted side.

7 Area of triangles

$A=\frac{1}{2}bh$: half a parallelogram.

Base 10, height 3: $A=15$



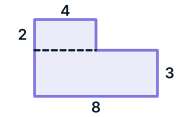
Watch out: don't forget the $\frac{1}{2}$.

8 Composite shapes

Split into shapes you know and add, or take a shape away.

L shape: $8\times3+4\times2=24+8=32$

10 by 6 with a 2 by 2 hole: $60-4=56$

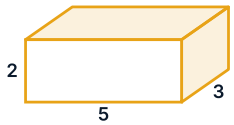


9 Volume of rectangular prisms

$V=lwh$, in cubic units.

$5\times3\times2=30$ cm³

Cube, side 3: $3\times3\times3=27$ cm³

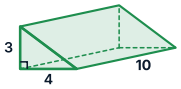


10 Volume of triangular prisms

$V=Ah$: area of the triangle times the length.

Triangle $\frac{1}{2}\times4\times3=6$, length 10: $V=60$

Triangle base 6, height 5: area 15. Length 4: $V=60$



11 Capacity

1 L = 1000 mL, 1 cm³ = 1 mL, 1 m³ = 1000 L.

250 mL = **0.25 L**

A 10 cm cube holds 1000 cm³, or **1 L**

1.5 L = **1500 mL**

12 Mass and temperature

1 t = 1000 kg, 1 kg = 1000 g.

2.5 kg = **2500 g**

Water freezes at 0°C and boils at 100°C

750 g = **0.75 kg**

From -2°C to 5°C is a rise of 7°